EE322 Optoelectronic Devices Fabrication Laboratory

Final Report

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Acknowledgments

Four months of course passed before I realized that I had to write this report[1]. Taking this course, *Optoelectronic Devices Fabrication Laboratory* , is a unique experience that combines the knowledge of semiconductor physics and the practical skills of semiconductor device fabrication. As a perspective graduate student who will continue the study in the field of flexible semiconductor devices, this course provides me with a valuable opportunity to practice the knowledge on the book and to understand the process of semiconductor device fabrication with experiment. This is the last report during my undergraduate journey, so this report can be regard as a mark of the end of my undergraduate study and the beginning of my graduate path. And I will memorize it as the last days in SUSTech.

I would like to express my sincere gratitude to Prof. Shuming Chen, the lecturer of this course, for his guidance and support throughout the course. He accepts me to join this course even though I have not taken the prerequisite course. The most valuable and fruitful part is the English writing suggestions and comments Prof. Chen provided. His comments and suggestions are very helpful for me to improve the quality of the report and my English writing skills.

My appreciation also goes to Mr. Tianhong Zhou, the teaching assistant, whose technical acumen and patient guidance were invaluable during the experimental phase of the course. His dedication to ensuring that we understood each step of the process and his readiness to assist us through any challenges were truly commendable. His hands-on approach made the experimental sessions not just a practical exercise but also an enjoyable learning experience.

I would also like to thank my fellow classmates, whose camaraderie and collaborative spirit made this final semester of my undergraduate studies a memorable one. As we part ways to pursue our individual paths, I extend my best wishes for their success and happiness in their future academic and professional pursuits.

[1]This report is presented with the format requirements of UTokyo, I would like to thank the author of the template for providing this LaTeXtemplate.

Abstract

This final report is the summary for the course EE322 Optoelectronic Devices Fabrication Laboratory. This report will cover every experiment in the course, starting from the physical vapor deposition (PVD) such as sputtering and vacuum evaporation, to the multilayer optoelectronic device fabrication, from light-emitting device such as Organic Light Emitting Diode (OLED), Quantum Dot Light Emitting Diode (QLED) and Alternating Current Electroluminance (ACEL) to the photovoltaic device, Organic Photovoltaic (OPV). The report will also include the results and analysis of the experiments, as well as the discussion of the results.

The report will also include the conclusion and the future work of the course. Starts from the basic fabrication techniques, and followed by the fabrication process of every device.

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Chapter 1

Fabrication Techniques of Optoelectronic Devices

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1.1 Fabrication Conditions

1.1.1 Clean Room

Cleanrooms are highly controlled environments used in semiconductor manufacturing to maintain strict quality standards and minimize contamination. They feature specialized HVAC systems that continuously filter and circulate air, removing particles as small as 0.3 micrometers through HEPA filters. Personnel wear cleanroom suits and PPE to prevent contaminants from entering. The materials used in cleanroom construction, such as stainless steel and epoxy-coated surfaces, are easy to clean and do not release particles. Cleanrooms are classified based on the maximum allowable concentration of particles per cubic meter of air, with ISO standards ranging from Class 1 (cleanest) to Class 9 (least clean). An example of clean room is shown in Fig. 1.1.

Figure 1.1: The appearance of nested cleanroom.

To save budgets, high level cleanrooms are nested in lower level cleanrooms, such as a Class 1 cleanroom nested in a Class 2 cleanroom. Cleanrooms are essential for manufacturing optoelectronic devices, as even small particles can affect device performance and yield. They offer benefits such as maintaining product and research quality, preventing defects, ensuring sterility, and safeguarding sensitive materials. Cleanrooms provide controlled environments that enable industries to manufacture high-quality products, conduct sensitive research, and ensure safety. By implementing stringent protocols, specialized equipment, and cleanliness practices, cleanrooms minimize contamination and support excellence in various sectors.

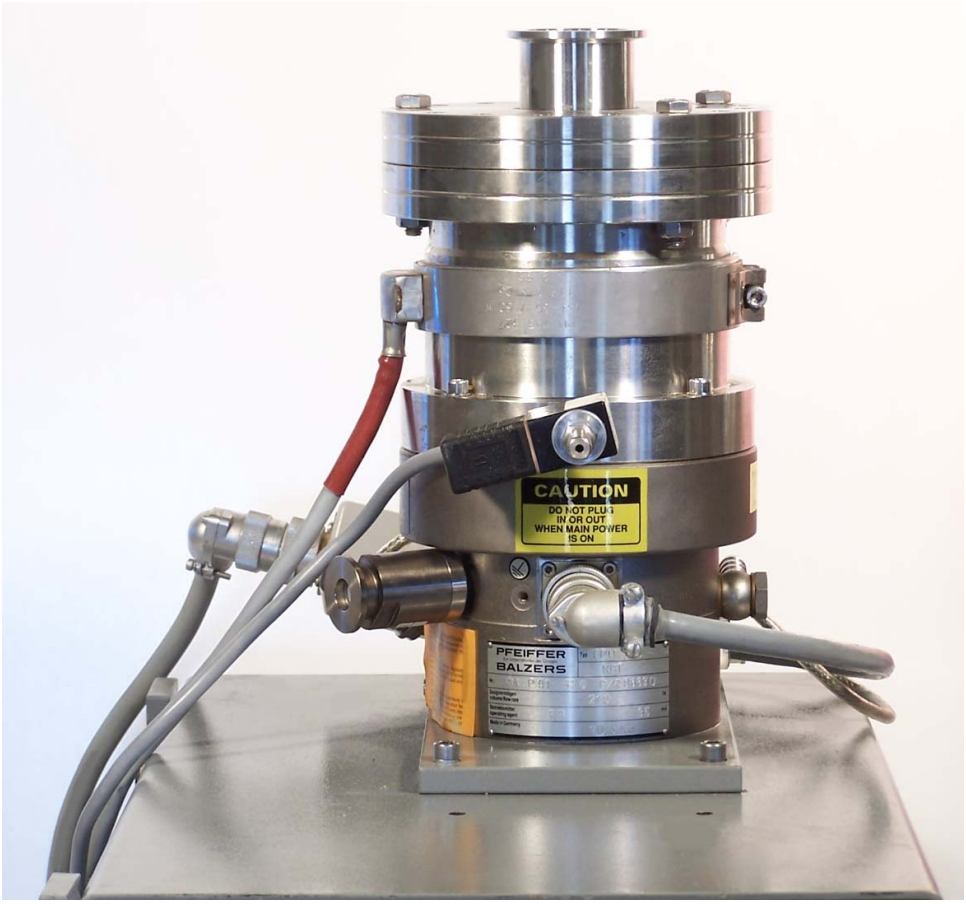

Figure 1.2: An example of the molecular pump.

1.1.2 High Vacuum

High vacuum is a common requirement in optoelectronic device fabrication. In most of our experiments, the devices are fabricated under the air pressure less than 10^{-3} torr. Specialized vacuum pumps (Fig. 1.2), such as turbomolecular pumps and cryogenic pumps, are employed to reduce pressure levels to the desired range. Leakage is also a critical concern in high vacuum systems, as even minute leaks can compromise the vacuum integrity. High vacuum is essential for processes such as physical vapor deposition (PVD), which involves depositing thin films of materials onto substrates. High vacuum systems feature vacuum pumps that remove air and other gases from the chamber, creating a low-pressure environment. This prevents gas molecules from interfering with the deposition process, ensuring uniform film thickness and quality. High vacuum systems are essential tools for producing optoelectronic devices with superior performance and reliability.

1.2 Fabrication Technology

1.2.1 Sputtering

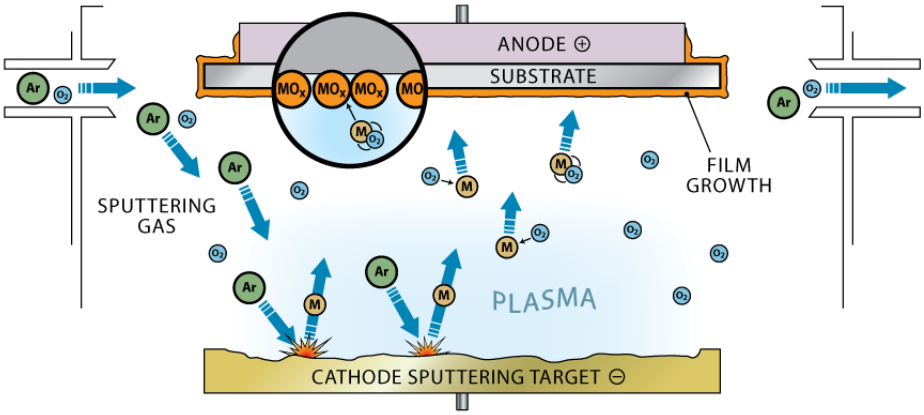

Figure 1.3: Illustration of the sputtering deposition process.

Sputtering technology is a type of physical vapor deposition (PVD) technology extensively utilized in thin film fabrication. The sputtering deposition process, as depicted in Fig. 1.3, necessitates a low-pressure (near vacuum) environment to ensure a high mean free path of the sputtered atoms. A high voltage is applied to the target material to generate plasma. The target material serves as the cathode, with the substrate positioned above the anode. The sputtered atoms will be deposited on the substrate to form the thin film. And we need a mask to control the shape of the thin film. In this experiment we use the tape as the mask.

1.2.2 Vacuum Evaporation

The vacuum evaporation system is shown in Fig. 1.4. The target materials are placed in the crucible (organic materials) or the tungsten boat (metal materials) in the vacuum chamber, and we can control which target material to be evaporated by the shutter.

When the system is working. The crucible or the tungsten boat is heated by the filament. And the target material will be evaporated and deposited on the substrate. The distance between the target and the substrate is crucial for the uniformity of the thin film. The mass deposition rate is given in Eq. (1.1).

$$R_m = C_m \left(\frac{M}{T}\right)^{\frac{1}{2}} \cos\theta \cos\varphi \frac{1}{r^2}\left(P_e(T) - P\right) \tag{1.1}$$

where C_m is a constant, M is the evaporant gram-molecular mass, r is the source-substrate distance, and $P_e(T)$ is the evaporant vapor pressure (a function of Temperature). The angle θ is the angle between the normal to the source surface and the direction of the evaporant vapor flux, and φ is the angle between the normal to the substrate surface and

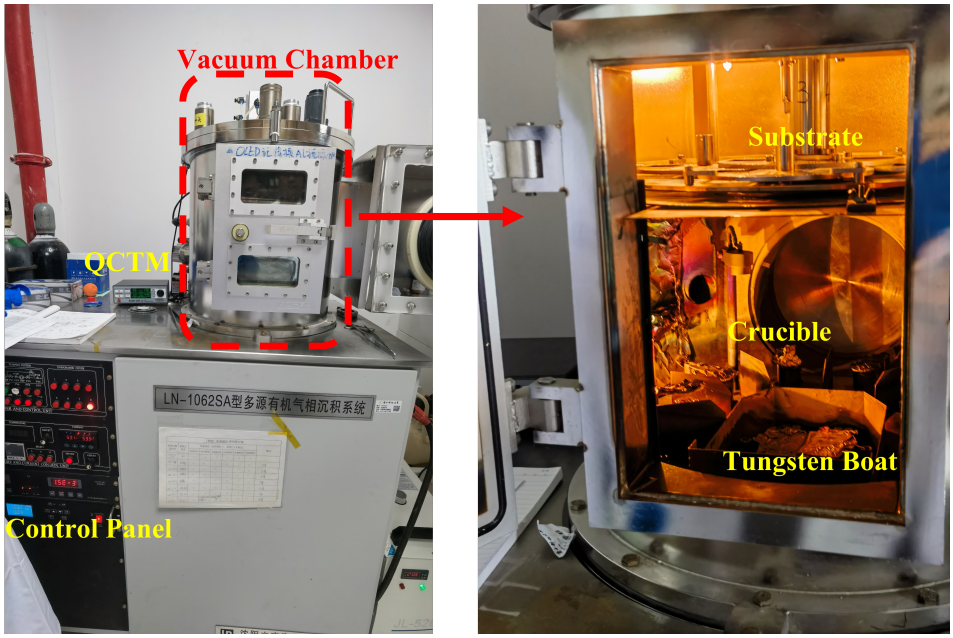

Figure 1.4: The vacuum evaporation system.

the direction of the evaporant vapor flux. If a substrate of width W is placed on the top of the source with distance r, and use ratio of the deposition rate between the center and the edge of the substrate to represent the uniformity of the thin film, we will have Eq. (1.2).

$$\sigma = 1 - \left(1 + \left(\frac{W}{2r}\right)^2\right)^{-2} \approx \frac{W^2}{2r^2} \tag{1.2}$$

The size of vacuum chamber is fixed, so according to Eq. (1.2) we can calculate the limits of uniformity for a given substrate size and source-substrate distance.

We can find a quartz crystal thickness monitor (QCTM) next to the vacuum chamber, it is used to measure the thickness and deposition rate of the thin film. It can reach a precision at the level of angstroms after calibration. The resonant frequency of the QCTM is given by Sauerbrey's equation in Eq. (1.3).

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{1.3}$$

If we consider the effective mass m is changed by Δm, the change of the resonant frequency Δf is given by Eq. (1.4)

$$\Delta f = f_1 - f_0 = -\frac{2\pi^2 f_0^3}{k}\Delta m \tag{1.4}$$

As a result, we can measure the thickness of the thin film by measuring the change of the resonant frequency. And we need to choose the material in the QCTM, so that we can transfer the mass of the thin film to thickness of the thin film.

1.2.3 Thin Film Patterning

Thin film patterning techniques, such as photolithography, laser direct writing, and screen printing, are essential processes in the fabrication of microelectronics, displays, sensors, and other electronic devices. These techniques enable the precise and controlled deposition of patterns onto thin films, allowing for the creation of complex structures and functional devices.

Photolithography is a widely used patterning technique that utilizes light-sensitive photoresist materials. A patterned mask is placed over the photoresist-coated substrate, and UV light is used to expose specific areas through the mask. The exposed regions undergo a chemical change, enabling selective removal of the photoresist during development. Photolithography offers high-resolution patterning capabilities and is a key process in semiconductor manufacturing, MEMS fabrication, and integrated circuit production. Laser direct writing employs focused laser beams to selectively ablate or modify thin films.

Figure 1.5: An example of laser direct writing pattern.

Fig. 1.5 shows the pattern created by laser direct writing. On the right side is *Cyberpunk 2077*, a famous video game character, and on the left side is *Bocchi the Rock!*, a popular Japanese anime character. By scanning the laser beam across the substrate, precise patterns can be created without the need for masks or photoresist. Laser direct writing allows for flexibility in pattern design and is particularly valuable for prototyping, research, and low-volume production. It finds applications in fields such as microelectronics, photonics, and flexible electronics.

Screen printing involves the use of a mesh screen with patterned openings. Ink or paste is applied to the screen, and a squeegee is used to force the ink through the open areas onto the substrate (Fig. 1.6). Screen printing is a versatile and cost-effective technique suitable for large-scale production of patterns on a variety of substrates. It is commonly used in the production of displays, circuit boards, photovoltaics, and biosensors.

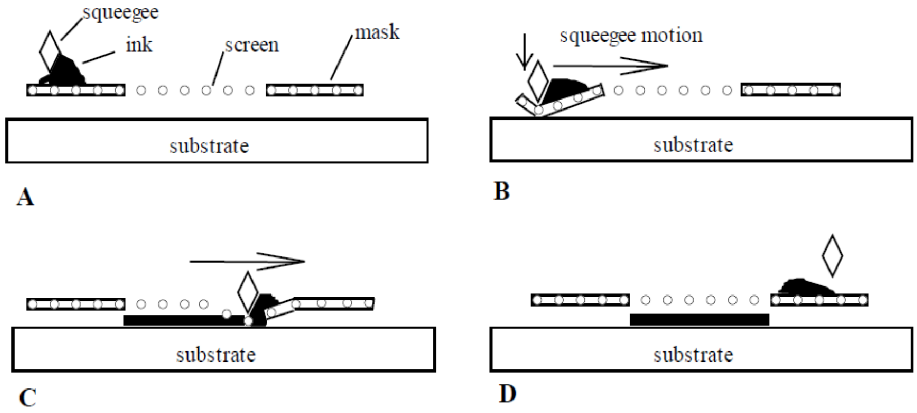

Figure 1.6: Illustration of the screen printing process.

1.2.4 Spin Coating

Spin coating is a widely used technique for depositing uniform thin films. The process of spin coating is shown in Fig. 1.7.

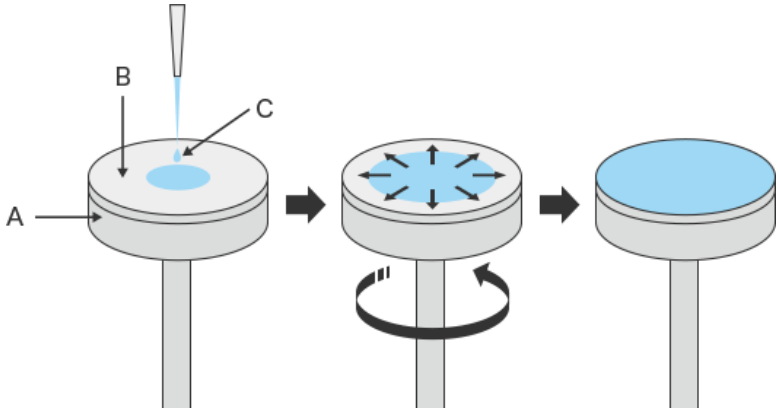

Figure 1.7: Illustration of the spin coating process.

Just as the name suggests, the substrate is spun at high speeds while a liquid precursor is dispensed onto the center of the substrate. Centrifugal force spreads the liquid across the substrate, forming a thin film. The rotation speed, dispensing rate, and viscosity of the precursor are critical parameters that influence the film thickness and uniformity. Spin coating is commonly used in the fabrication of thin films for microelectronics, displays, sensors, and photovoltaics. This technique offers excellent control over film thickness and quality, making it a versatile and essential process in thin film deposition.

1.2.5 Encapsulation

Encapsulation is a critical step in the fabrication of optoelectronic devices. Oxygens and moisture in the air can degrade the performance and lifetime of devices, especially organic optoelectronic devices. Encapsulation materials, such as glass, metal, and polymers, are used to protect devices from environmental factors.

Chapter 2

Fabrication Experiment of Optoelectronic Devices

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2.1 Indium Zinc-Oxide Electrode

2.1.1 Introduction

The journey of EE322 started with the fabrication experiment of Indium Zinc-Oxide (IZO) electrode. IZO is a transparent conductive oxide (TCO) material that is widely used in optoelectronic devices, it can be find in almost all the rest of the experiments in this course. The IZO electrode is fabricated by sputtering technology introduced in Section 1.2.1. It is crucial for light-emitting and photovoltaic devices to have a transparent electrode, as it allows light to pass through and reach the active or functional layer.

Table 2.1: Recipes and result of IZO deposition

Recipe Index	**Air Pressure ($\times 10^{-1}$ Pa)**	**O_2 Gas Flow (sccm)**	**Thickness (nm)**	**Resistance ($\Omega/\square$)**
1	2.0	0	36.6	259.8
2	3.0	0	26.7	330.8
3	5.0	0	33.0	491.2
4	7.0	0	28.5	4049.3
5	3.0	1	28.2	512.3
6	3.0	2	27.6	1905.4
7	3.0	3	31.4	Too large

During the fabrication, the gas flow of O_2 and the air pressure in the vacuum chamber are adjusted for each film. The O_2 gas flow ranges from 0 sccm to 3 sccm, while the air pressure is maintained between 2.0×10^{-1} Pa and 7.0×10^{-1} Pa. The deposition time is fixed at 8 minutes. 7 recipes were tested, and two samples were fabricated parallel using the same recipe to ensure experimental reliability. The recipes and their corresponding mechanical and electrical properties are presented in Table 2.1.

Sheet resistance was measured using the four-point probe method with a current set to 45.32 μA, at 5 different points (center and 4 corners). The data in the table represents the average value of 10 points (5 points for each parallel sample). Thickness was measured using a step profiler at 3 different points (randomly selected), with the data in the table reflecting the average value of 6 points (3 points for each parallel sample). Transmissivity was measured using a spectrometer, with an integrating sphere utilized to enhance the measured light intensity.

2.1.2 Results and Discussion

a. The effect of air pressure

In this section, the impact of air pressure on the resistance and thickness of IZO thin films is investigated by comparing recipes 1, 2, 3, and 4. These thin films were fabricated under 0 sccm O_2 gas flow but with varying air pressures. The experimental results for film thickness are illustrated in Fig. 2.1. It is evident that the thickness of the IZO thin film exhibits a decreasing trend as the air pressure increases. This phenomenon can be attributed to the influence of air pressure on the mean free path of sputtered atoms. At higher air pressures, the mean free path of sputtered atoms is reduced due to collisions

with air molecules. Consequently, fewer atoms are deposited on the substrate, leading to a decrease in the thickness of the IZO thin film.

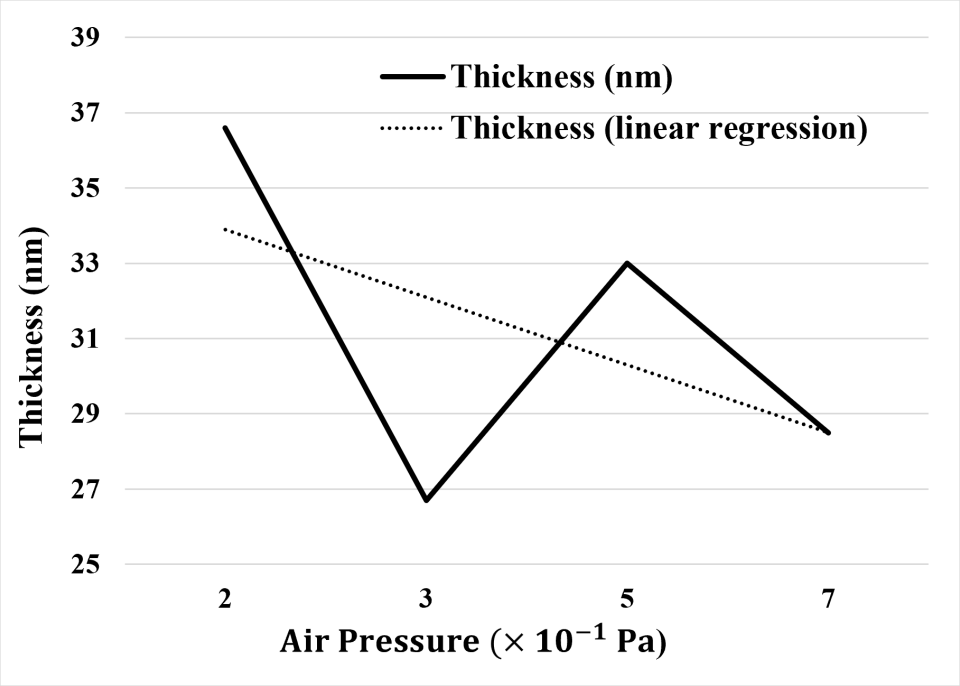


Figure 2.1: The thickness of films under different air pressure.

And the result of sheet resistance is shown in Fig. 2.2. The sheet resistance shows an increasing trend with the increase of air pressure. In the experiment, we used the substrate of the same size. As a result, the sheet resistance, R_s can be calculated by:

$$R_s = \rho/h \tag{2.1}$$

Where ρ is the resistivity, and h is the thickness of the film. This result matches the trend of the thickness in general.

As for the transmissivity, the raw data is shown in Fig. 2.3(a), and we can observe that the data oscillates severely. Most part of the data overlaps with each other, we could only observe all films have a transmissivity of around 90% in the visible spectrum. As a result, a moving average filter (WIN_SIZE = 30) is applied to remove the outliers (whose value is not in the domain of [WIN_AVG-2×WIN_VAR, WIN_AVG+2×WIN_VAR]), and interpolate the missing data with linear interpolation. The cleaned data is shown in Fig. 2.3(b), revealing a slight increase in transmissivity with higher air pressure, albeit not markedly significant. This trend can be primarily attributed to the reduction in the thickness of the IZO thin film.

b. The effect of O_2 gas flow

By comparing the recipe 2, 5, 6, 7 the effect of O_2 gas flow on the characterization of the IZO thin film can be observed. These films were fabricated under 3.0 $\times 10^{-1}$ Pa air pressure but with different O_2 gas flow. The experiment result of thickness is shown

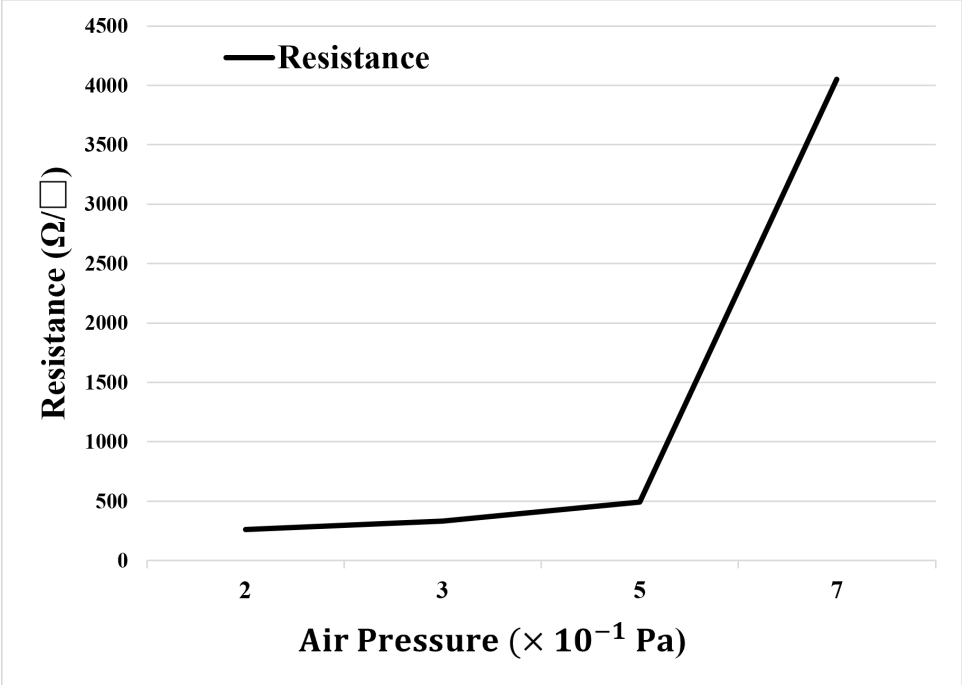
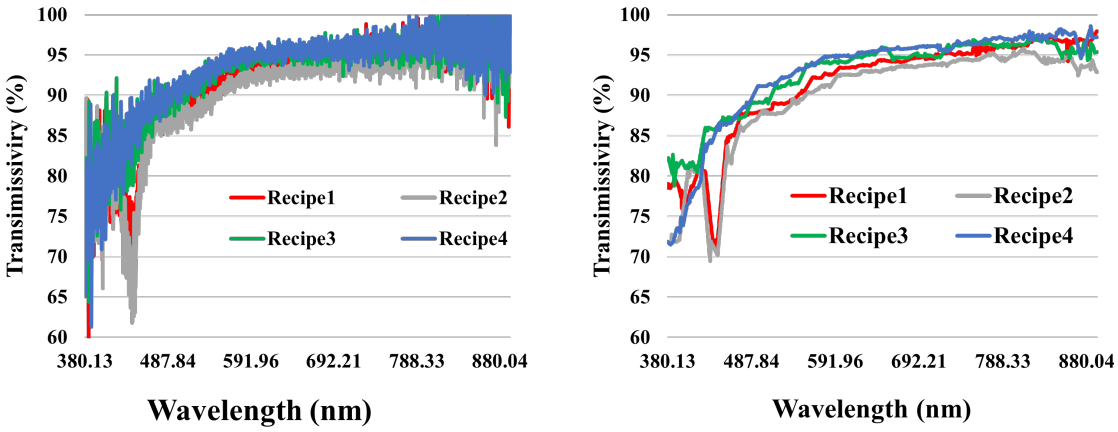


Figure 2.2: The resistance of films under different air pressure.



(a) (b)

Figure 2.3: The transmissivity of films under different air pressure.

in Fig. 2.4. And an increasing trend of the thickness with the increase of O_2 gas flow can be observed. This trend may be attributed to the role of adequate O_2 gas flow in promoting the formation of IZO thin films with improved crystallinity and accelerating the deposition process, consequently leading to an increase in film thickness.

The result of sheet resistance is shown in Fig. 2.5. The sheet resistance also shows an increasing trend with the increase of O_2 gas flow. A comparison between recipe 3 (air pressure = 5.0 $\times 10^{-1}$ Pa, O_2 gas flow = 0 sccm) and recipe 7 (air pressure = 3.0 $\times 10^{-1}$ Pa, O_2 gas flow = 3 sccm) reveals that although recipe 3 is thinner than recipe 7, its resistance

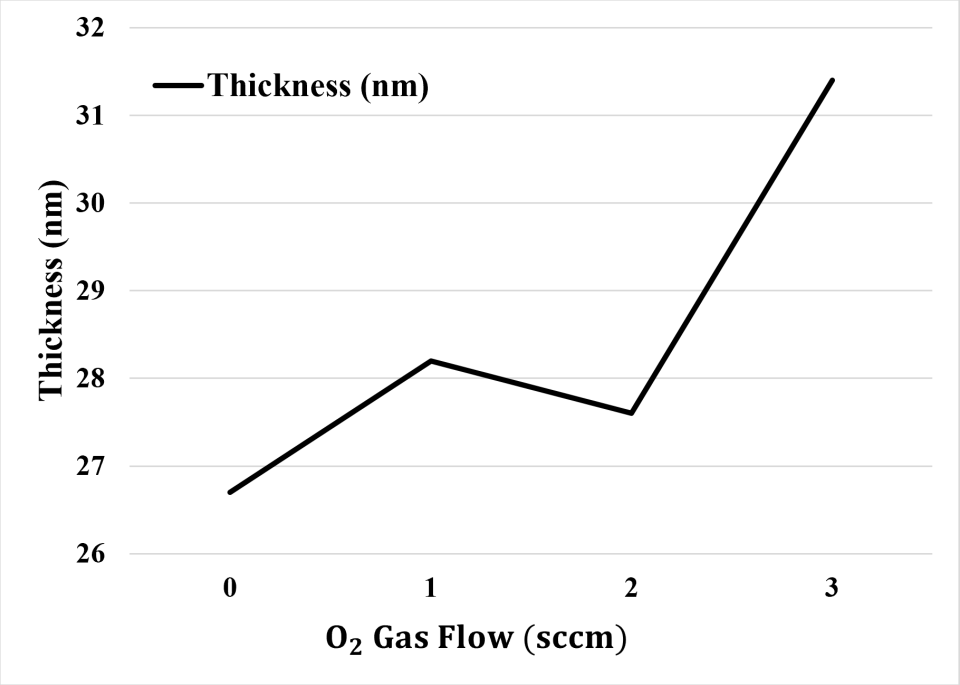


Figure 2.4: The thickness of films under different O_2 gas flow.

is significantly lower. This difference can be attributed to the higher crystallinity of the IZO thin film fabricated under 3 sccm O_2 gas flow, resulting in reduced conductivity and hence higher resistance.

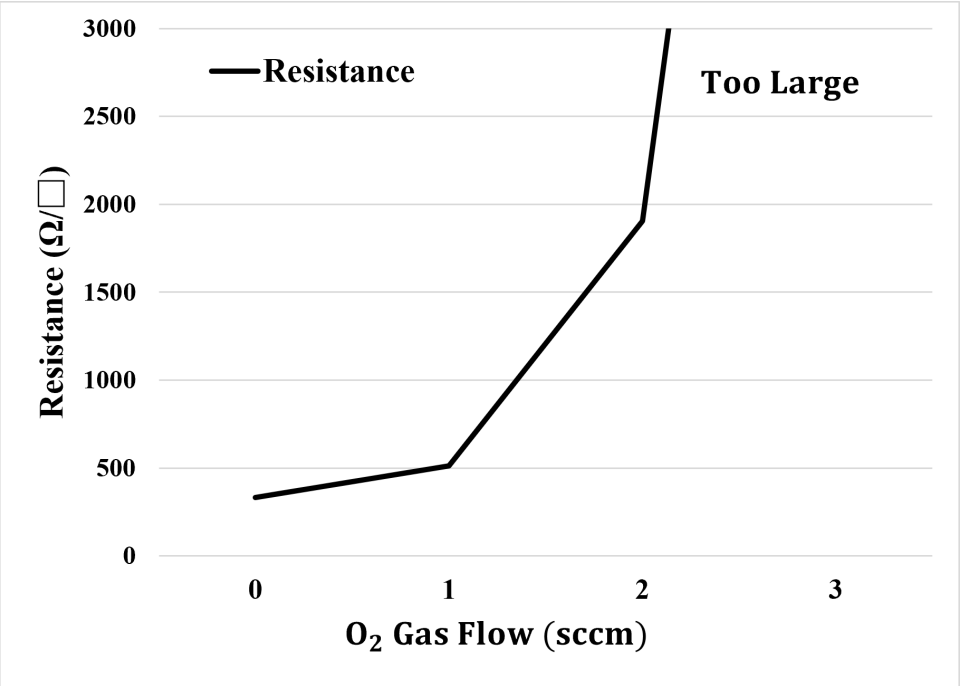
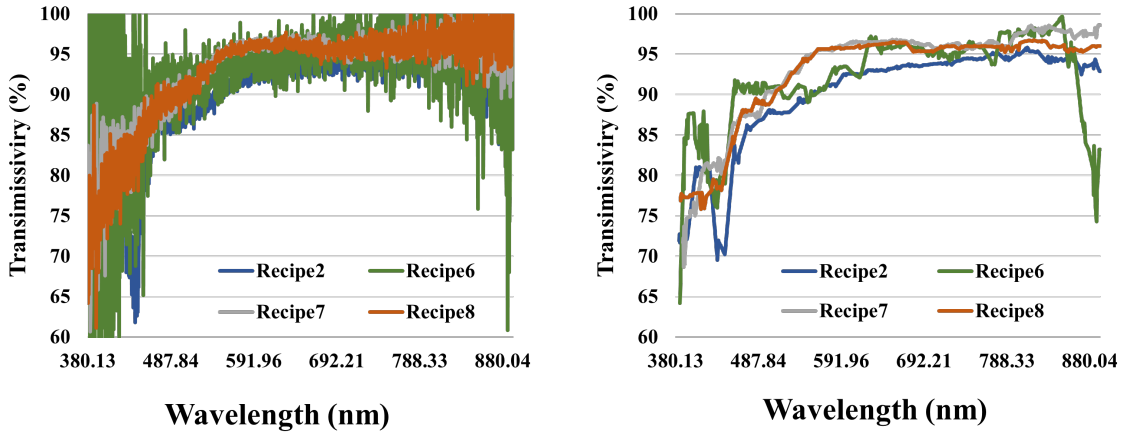


Figure 2.5: The resistance of films under different O_2 gas flow.

The transmissivity data of the films were processed using the same method as described in Section a., and the results are depicted in Fig. 2.6. All four films exhibit a transmissivity of approximately 90% in the visible spectrum, with no significant or consistent trend observed with varying O_2 gas flow rates.



(a) (b)

Figure 2.6: The transmissivity of films under different O_2 gas flow.

2.2 Thin Film Vacuum Evaporation

2.2.1 Introduction

One of the significant shortages of the magnetron sputtering technology is that its disability to deposit the organic compounds. The high energy ions in the plasma could break the molecular bonds, leading to the fragmentation of the organic molecules.

To deposit these sensitive and fragile materials, the vacuum evaporation technology is employed. The basic principle and the system's structure are introduced in Section 1.2.2. In the experiment, six different samples were produced in this experiment. Two different target materials were used and maximum three layers were deposited on the substrate. Sample 1 is intended to deposit 100 nm Ag. Sample 2 is 100 nm Alq. Sample 3 is a stack of Ag 100 nm / Alq 100 nm / Ag 20 nm. Sample 4 is a stack of Ag 100 nm / 150 nm / Ag 20 nm. Sample 5 is 20 nm Ag. And Sample 6 is a stack of Ag 20 nm / Alq 40 nm. The structure of all the samples are shown in Fig. 2.7. Alq used here is a kind of organic

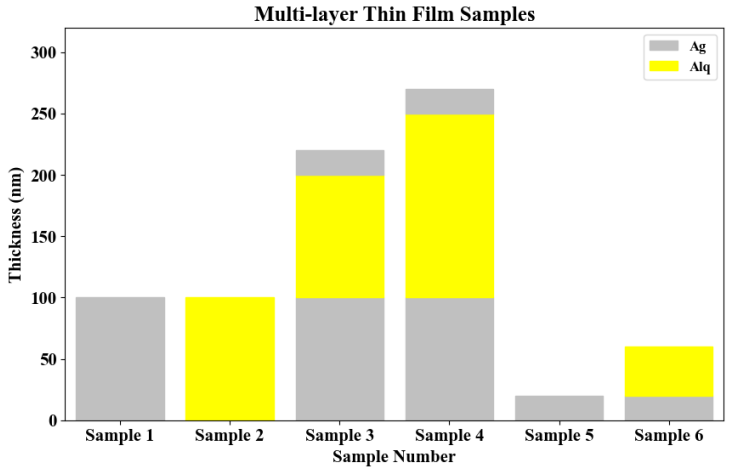


Figure 2.7: The structure of six thin films fabricated by vacuum evaporation.

compound, and it is used as the hole-transporting layer in OLED. The structure of Alq is shown in Fig. 2.8.

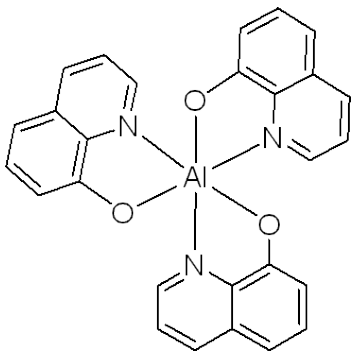


Figure 2.8: The structure of Alq.

Different from the sputtering technology, we can take out the substrate once the one

samples is finished. It would take a lot of time to take out the samples and recreate the vacuum environment in this experiment. Instead, we can use the mask to cover the substrate and deposit different layers. When all the samples are finished, we can take out the substrate. And the fabrication process is:

- Ag 100 nm, for Sample 1, 3, 4
- Alq 100 nm, for Sample 2, 3, 4
- Alq 50 nm, for Sample 4
- Ag 20 nm, for Sample 5, 6
- Alq 40 nm, for Sample 6

The thickness is monitored by the QCTM, and the deposition rate is controlled to a constant value.

2.2.2 Results and Discussion

The final samples are shown in Fig. 2.9. Different colors and reflectivity can be observed.

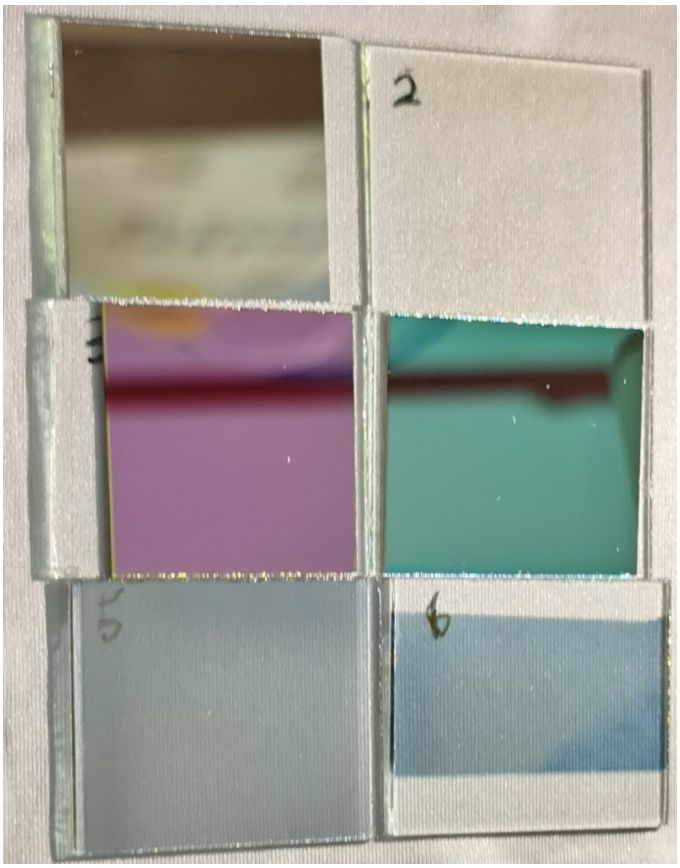


Figure 2.9: Photo of the samples.

a. Compression of Ag and Alq

When testing reflectivity of Sample 1, we found the reflectivity is 110%, which means it has a higher reflectivity compared with the standard white board. As a result, the reflectivity of Sample 1 is recognized as 100% in the following testing. The reflectivity and transmittance of Sample 2 are shown in Fig. 2.10.

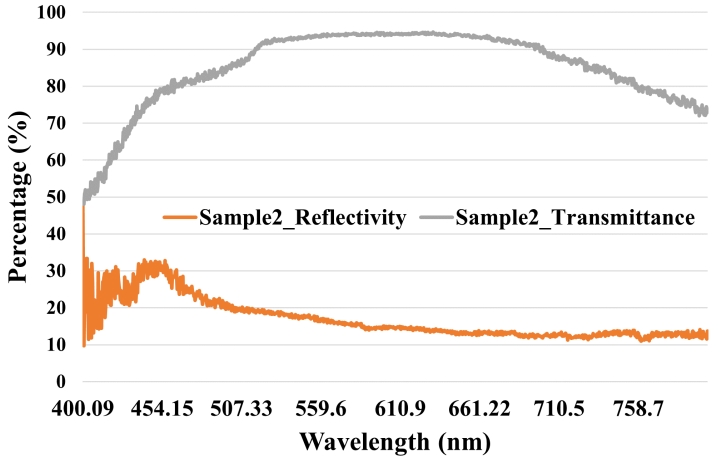
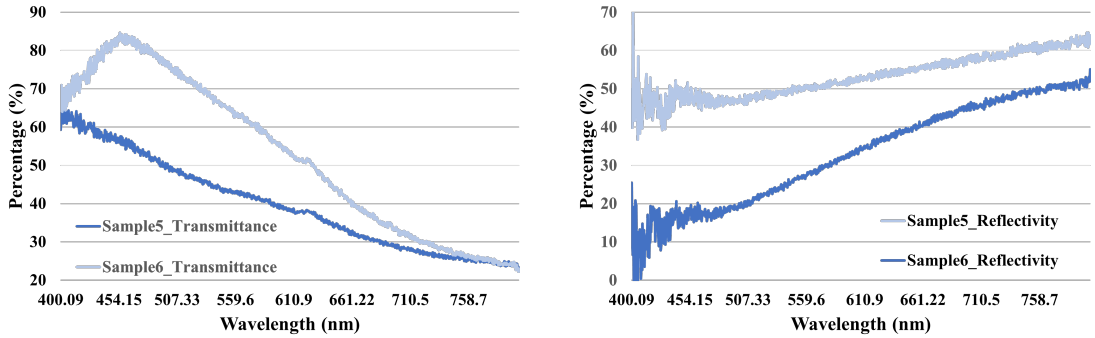


Figure 2.10: The reflectivity and transmittance of Sample 2.

And we could find that Alq is highly transparent in the visible light region. And that is the reason why it is suitable for the hole-transporting layer in OLED.

b. Increasing transmittance by Alq

Alq can also increase the transmittance of the thin silver film. The transmittance and reflectivity of Sample 5 and Sample 6 are shown in Fig. 2.11. This result also allies with the observed color of the samples shown in Fig. 2.9.



(a) (b)

Figure 2.11: Effect of adding another layer of Alq on Ag.

c. Effect of the thickness of Alq

Sample 3 and Sample 4 are designed to compare the effect of the thickness of the Alq layer. The reflectivity of Sample 3 and Sample 4 are shown in Fig. 2.12. The lowest reflectivity is observed at about 510 nm when the thickness of Alq is 100 nm, which means green is missing from the reflected light, leading to the color of bluish violet. And the lowest reflectivity is observed at about 660 nm when the thickness of Alq is creased to 150 nm, which means red light is missing, and it will show the color of bluish green. The result shows that the thickness of Alq is crucial for the performance of the thin film.

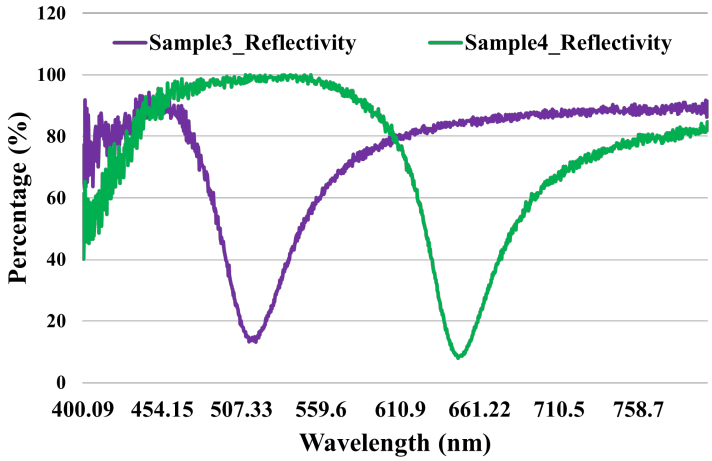


Figure 2.12: Effect of the thickness of Alq.

d. Thickness Verification

QCTM is used to monitor the thickness during the fabrication. Here, we are going to use another three methods to verify the thickness of the thin film. The involved methods are:

- d1: Calculate by sheet resistance.
- d2: Simulate with n-k database.
- d3: Measure with step profiler.

The sheet resistance can only work with one layer and conductive thin film. The resistance can be calculated by Eq. (2.2).

$$R = \frac{\rho}{t} \tag{2.2}$$

where R is the sheet resistance, ρ is the resistivity, and t is the thickness. But this method is not suitable for multi-layer thin films. As for the simulation method, the given n-k database and LabVIEW software are used. The resistivity ρ of Ag is 16 $\Omega \cdot nm$. We can calculate the thickness of Sample 1 is 53.3 nm, and the thickness of Sample 5 is 5.3 nm. All the results are shown in Table 2.2.

Table 2.2: Thickness of the thin film fabricated by vacuum evaporation (nm).

Sample Index	**d1 (Calculation method)**	**d2 (Simulation method)**	**(d3 Measurement method)**	**QCTM**
1	53.3	-	77.4	100
2	-	-	86.7	100
3	-	214	-	220
4	-	257	-	270
5	5.3	-	27.6	20
6	-	-	56.1	60

We can find that the result of simulation is pretty close to the result of QCTM. And there is a roughly 10% difference between the result of QCTM and the result of step profiler. This result indicates that the fabrication is successful.

However, the result by calculation is far from the rest of results. In my opinion, this error may be caused by the non-uniformity of the thin film. And the assumption of regard it as a uniform and large thin film is not suitable for the real situation.

2.3 Organic Light Emitting Diode

2.3.1 Introduction

Having mastered the basic fabrication techniques, we are now ready to move on to more complex devices. The Organic Light Emitting Diode (OLED) is a type of light-emitting device that utilizes organic compounds to emit light when an electric current is applied. This technology boasts advantages over conventional displays, including enhanced image quality, true black levels, higher contrast ratios, and faster refresh rates. Furthermore, OLEDs enable innovative design possibilities due to their flexibility, thinness, and compatibility with various substrate materials. Despite challenges like degradation and burn-in, continuous research is addressing these issues and improving OLED performance. In essence, OLEDs are a transformative technology with substantial potential in display and lighting applications.

Device Structure

The OLED device is composed of several key layers: Hole Injection Layer (HIL), Hole Transport Layer (HTL), Emission Layer (EML), Electron Transport Layer (ETL), and Electron Injection Layer. (Each layer plays a distinct role.) The structure of the OLED device is illustrated in Figure 2.13. Unlike the conventional LED device, which is characterized by fewer layers and a simpler design, the OLED device necessitates a more elaborate arrangement of multiple layers to achieve optimal light emission performance.

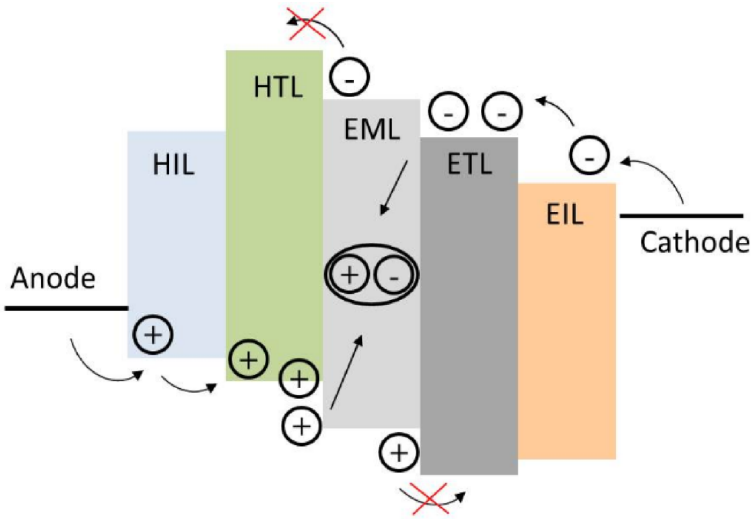


Figure 2.13: The structure of OLED.

In this experiment, all devices employ ITO (indium tin oxide) as the anode and aluminum (Al) as the cathode. The specific material and thickness specifications for each device are detailed in Table 2.3.

All four devices primarily differ in their (Electron-Migration Layer, EML). Device 1 and Device 3 can be employed to examine the impact of material choice, while Devices 2 to 4 facilitate a discussion on the effect of the (Electron Transport Layer, ETL) thickness.

Table 2.3: The composition of the fabricated OLED devices.

Device	HIL	HTL	EML	ETL	EIL
1	HAT-CN:20 nm	NPB:40 nm	Alq_3:20 nm	TPBi:40 nm	LiF:1 nm
2	HAT-CN:20 nm	NPB:40 nm	CBP:$Ir(ppy)_3$:20 nm	TPBi:30 nm	LiF:1 nm
3	HAT-CN:20 nm	NPB:40 nm	CBP:$Ir(ppy)_3$:20 nm	TPBi:40 nm	LiF:1 nm
4	HAT-CN:20 nm	NPB:40 nm	CBP:$Ir(ppy)_3$:20 nm	TPBi:50 nm	LiF:1 nm

Materials and its function

In this experimental study, we employ several novel materials, and their detailed inventory is as follows:

- **HAT-CN**, or 1,3,5-Triazine-2,4,6-triyltribenzonitrile, is a specialized hole-transporting material (HTM) characterized by high hole mobility and well-matched energy levels. This compatibility facilitates efficient charge injection and transport within the OLED device.
- **NPB**, or N,N'-Di(naphthalen-1-yl)-N,N'-diphenyl-benzidine. NPB is a high-performance hole-transporting material, exhibits exceptional hole mobility and suitable energy levels. Its alignment with the adjacent layers in OLED devices facilitates efficient charge injection and transport.
- **Alq_3**, or Tris(8-hydroxyquinolinato)aluminum. Alq3 acts as a phosphorescent emitter, directly contributing to light emission in some OLED devices.
- **CBP:Ir(ppy)$_3$**, or 4,4'-Bis(N-carbazolyl)-1,1'-biphenyl (a host material), plays a crucial role in organic light-emitting diodes (OLEDs). It serves as the host for the emissive layer and incorporates $Ir(ppy)_3$, a phosphorescent dopant. The combination of CBP:$Ir(ppy)_3$ enhances device performance by enabling efficient charge transport within the emitting layer and functioning as a phosphorescent emitter. This leads to notable improvements, such as higher external quantum efficiency, lower operational voltage, and longer device lifetimes, compared to OLEDs that solely rely on fluorescent emitters.
- **TPBi**, or 2,2',2"-(1,3,5-Benzinetriyl)-tris(1-phenyl-1-H-benzimidazole), is a widely employed electron-transporting material. Its exceptional electron mobility, thermal stability, and well-matched energy levels with the adjacent layers in an OLED device facilitate efficient charge injection and transport (electron transfer).

2.3.2 Results and Discussion

a. External quantum efficiency (EQE)

Regarding light-emitting devices, the parameter of utmost concern is the External Quantum Efficiency (EQE), which is defined as:

$$EQE = \frac{\text{Outcoupled Photons: } N(p)}{\text{Injected Electrons: } N(e)} \tag{2.3}$$

The photodiode was employed for measuring the photon current (I), which could be determined through a calculation.

$$I = x \cdot \frac{\int S(\lambda)R(\lambda)d\lambda}{\int S(\lambda)d\lambda} \tag{2.4}$$

where x represents the power of the outcoupled photon, $S(\lambda)$ is the normalized spectral distribution of the device (also known as the spectral power density), and $R(\lambda)$ is the photodiode's response curve, typically provided by the manufacturer. To measure I, a V-I (voltage-current) device is employed. $S(\lambda)$ can be obtained through spectral measurement techniques, while $R(\lambda)$ serves as a given input. By utilizing Eq. 2.4, we can calculate the photon power x. Integrating Eq. 2.4 with Planck's law enables us to determine the number of outcoupled photons (denoted as $N_{photons}$).

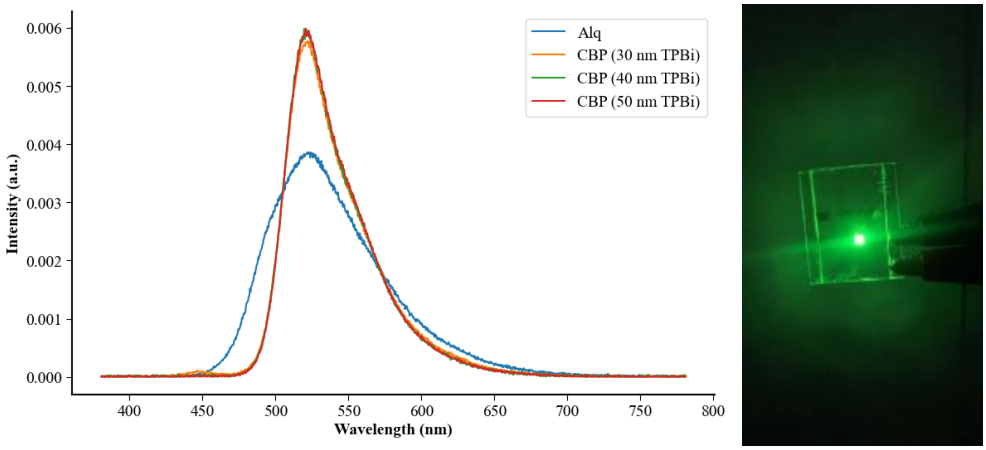
$$N(p) = \int \frac{x\frac{S(\lambda)}{\int S(\lambda)d\lambda}}{hv} d\lambda \tag{2.5}$$

The number of injected electrons can be determined using the following equation:

$$N(e) = \frac{I}{e} \tag{2.6}$$

In this context, where e represents the elementary charge, by integrating Eq. (2.5) and Eq. (2.6), we can derive the external quantum efficiency (EQE) of the device.

b. Spectrum



(a) (b)

Figure 2.14: (a) Spectrum of the four devices. (b) Optical appearance.

Due to fabrication discrepancies, the data from sample devices was subjected to testing. The device's spectral characteristics, as depicted in Fig. 2.14, showed a broad range from approximately 450 to 650 nm. It should be noticed that Fig. 2.14 shows the normalized spectrum, which means the area below the curve is 1. The optical appearance (Fig. 2.14(b)) exhibited green light, in line with the spectral findings. Notably, the light emitted by the CBP:Ir(ppy)$_3$ devices appeared purer and brighter compared to those of the Alq devices.

c. J-V Characterization

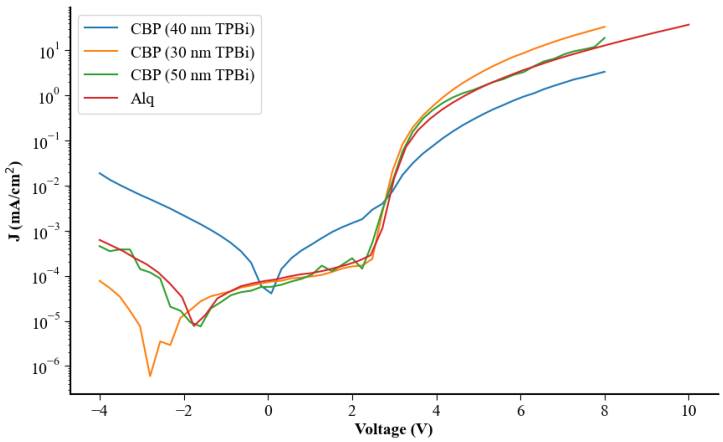


Figure 2.15: J-V characterization.

The J-V characteristics of the devices are measured using a Source Measure Unit (SMU). By varying the voltage from -4 V to 10 V, the current flowing through the device is recorded, as depicted in the data. Among the three CBP:Ir(ppy)_3 devices, it is observed that the resistance rises as the thickness of the electron transport layer increases. The turn-on voltage for all the devices is approximately 3 V.

d. Luminous

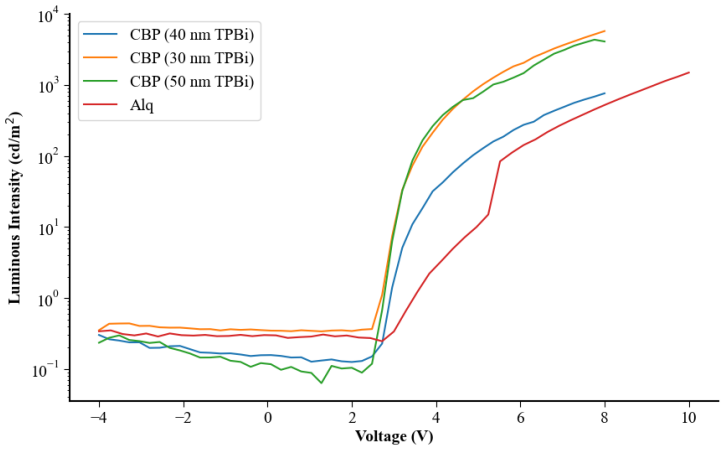


Figure 2.16: Luminous-voltage characterization.

The relationship between luminosity and spectral properties is depicted in Equation 2.7.

$$Lm(\lambda) = S(\lambda) \cdot Vis(\lambda) \tag{2.7}$$

$Vis(\lambda)$, often referred to as the human eye's response curve, is specified by the CIE. The luminous intensity of the four devices under varying voltages is depicted in Figure 2.16. At

the same voltage, the luminous output of the CBP:Ir(ppy)$_3$ device is significantly greater than that of the Alq devices, indicating its higher efficiency in capturing both singlet and triplet excitons.

e. EQE result

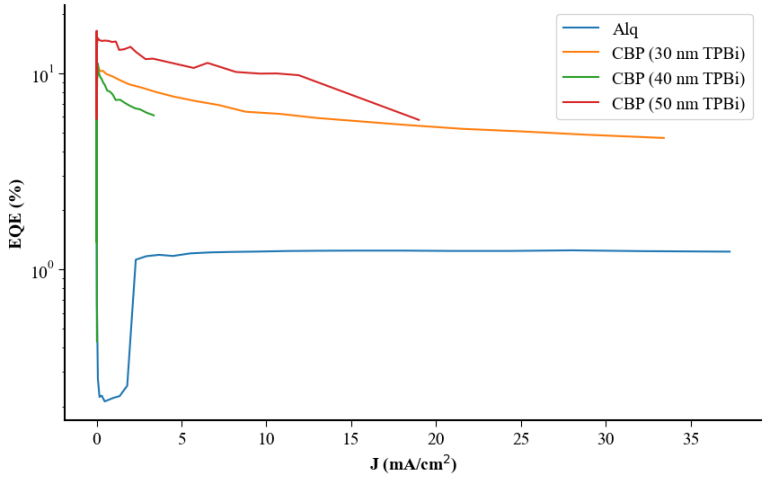


Figure 2.17: EQE.

The EQE of the devices is presented in Fig. 2.17. The CBP:Ir(ppy)$_3$ devices exhibit a higher external quantum efficiency (EQE) compared to the Alq devices. Notably, adjusting the thickness of the electron transmission layer allows for optimizing the EQE.

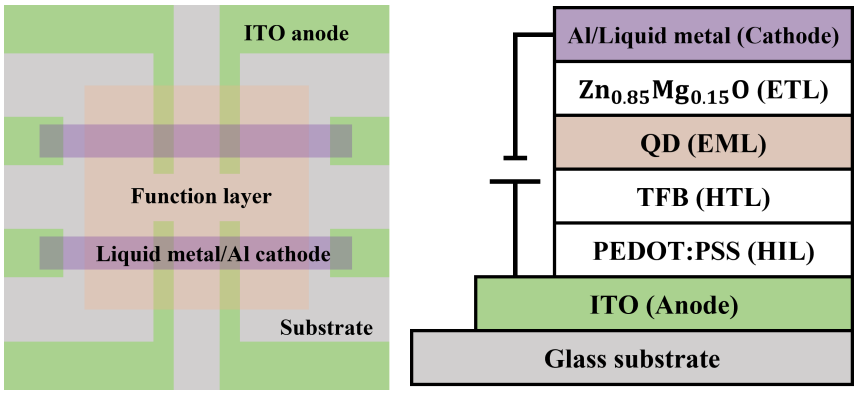
2.4 Quantum-Dot Light Emitting Diode

2.4.1 Introduction

Quantum-dot Light-Emitting Diodes (QLEDs), a revolutionary display technology, have garnered significant attention in recent years. It offers better color purity, higher brightness, and lower power consumption compared to traditional LCDs and OLEDs. QLED displays integrate a quantum dot (QD) layer between the backlight and the liquid crystal layer in an LCD panel, combining the benefits of both technologies.

The most important part of the QLED devices is the emitting layer, which is introduced to emit light and determine the color of the light. Based on the color of light, th QD materials are divided into R-QD (red), G-QD (green), and B-QD (blue). QDs are nanoscale semiconductor particles whose electronic properties are profoundly influenced by their size, which typically ranges from 2 to 10 nanometers. These properties derive from the quantum confinement effect, where the reduction in particle size leads to discrete, quantized energy levels. To emit light, the QDs are excited by an external energy source, such as an electric field or light. When the QDs return to their ground state, they release energy in the form of photons. The color of the emitted light is determined by the size of the QDs. The size-dependent emission characteristic of quantum dots is a consequence of quantum confinement. Smaller quantum dots have a larger bandgap, resulting in higher energy emissions (blue light), while larger quantum dots have a smaller bandgap, emitting lower energy (red light). The precise control of quantum dot size during synthesis allows for the tuning of emitted light across the visible spectrum.

Device Structure



(a) (b)

Figure 2.18: The structure of the device. (a) Top View (b) Side View.

The structure of the fabricated devices is shown in Fig. 2.18. It can be observed that each substrate contains four pixels. The device contains seven layers, including the substrate, anode, hole injection layer (HIL), hole transmission layer (HTL), emitting layer (EML), electron transmission layer (ETL), and cathode. The materials used for each layer will be introduced in the following section.

Materials and its function

In this experiment, several novel materials were employed, and their detailed inventory is as follows:

1. **PEDOT:PSS** or Poly(3,4-ethylenedioxythiophene) polystyrene sulfonate is a conductive polymer that serves as the hole injection layer. It is a transparent, conductive polymer that is commonly used in flexible electronics devices.

2. **TFB** or Poly[(9,9-dioctylfluorenyl-2,7-diyl)-co-(4,4'-(N-(4-sec-butylphenyl)) diphenylamine)] is a hole transport layer that is used to transport holes from the anode to the emitting layer.

3. $\mathbf{Zn_{0.85}Mg_{0.15}O}$ is an electron transport layer that is used to transport electrons from the cathode to the emitting layer.

Fabrication Process

Table 2.4: Condensed Summary of Fabricated QLED Samples

Sample Index	Environment	HTL (8mg/mL TFB, 50uL)	EML	ETL ($Zn_{0.85}Mg_{0.15}O$ 30mg/mL, 50uL)
1, 7, 4-6	1, 4-6: In glove box 7: Out of glove box	3000rpm, 40s, 100°C, 20min	R-QD 15mg/mL, 35uL, 100°C, 5min	1,8: 3000rpm, 40s, 100°C, 10min 4: 1500rpm, 5: 2000rpm, 6: 4000rpm
2, 8	2: In glove box 8: Out of glove box	3000rpm, 40s, 100°C, 20min	G-QD 10mg/mL, 35uL, 100°C, 5min	3000rpm, 40s, 100°C, 10min
3, 9	3: In glove box 9: Out of glove box	3000rpm, 40s, 100°C, 20min	B-QD 10mg/mL, 35uL, 100°C, 5min	3000rpm, 40s, 100°C, 10min

In this experiment, the HIL has been dep[osited on the ITO substrate before the laboratory. And two different fabrication methods were employed to complete the rest of the device structure. The first method involved vacuum evaporation technology to fabricate the cathode, while the second utilized spin coating to deploy it, and the liquid metal was taken the place of the Al cathode. The information of each layer of every sample is shown in Table 2.4.

2.4.2 Results and Discussion

a. Spectrum

The optical appearance of the devices is shown in Fig. 2.19.

The spectrum of RGB-QD devices is shown in Fig. 2.20. The spectrum of QLED has a sharp pulse in Red, Green and Blue. And the peak wavelength of the devices is indicted in Fig. 2.20 with dotted lines.

It can be observed that the wavelength of the peak intensity of red pixel is 628 nm, the green pixel is 535 nm, and the blue pixel is 474 nm. The narrow bandwidth of the spectrum indicates the high color purity of the devices.

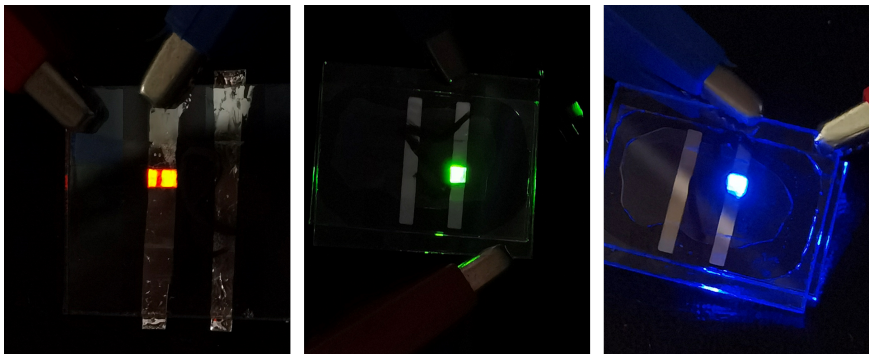


Figure 2.19: The optical appearance of the devices.

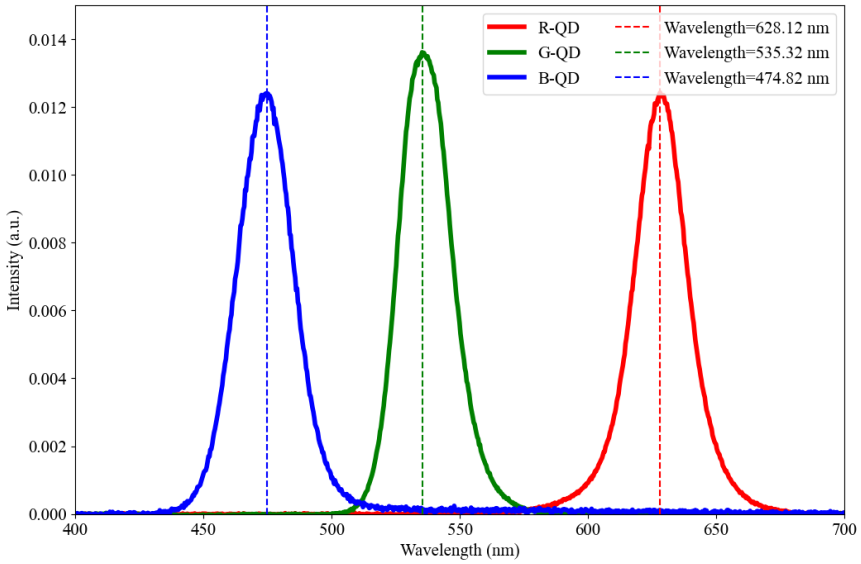


Figure 2.20: The spectrum of RGB-QD devices.

b. Luminous

The luminous of the four devices under different voltage is shown in Fig. 2.21. It can be observed that the R-QD has the highest turn on voltage which is around 2 V. And the G-QD has the lowest turn on voltage but the highest luminous.

And it can also be observed that the best speed for spining the ETL is 2000 rpm. High speed will result in a thinner ETL layer, and it will increase the leakage current of the device, resulting in a higher power consumption.

c. J-V Characterization

The J-V characterization of the devices is tested with a SMU(source measure unit). Under an arrangement of voltage from -4 V to 6 V. The current density of the devices is shown in Fig. 2.22.

According to Fig. 2.22, the current density of the devices increases with the voltage. And R-QD, G-QD and B-QD have a similar J-V curve. And if we change the thickness of the ETL, the J-V curve will change greatly. Sample 6 with the thinnest ETL tolerates

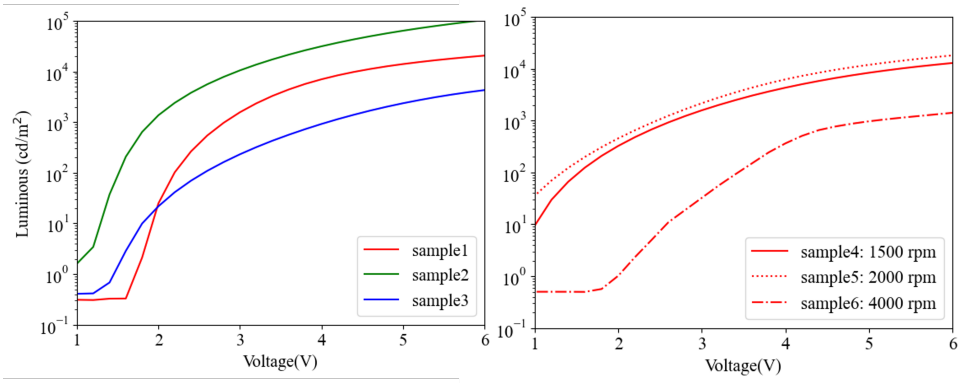


Figure 2.21: The luminous of the devices under different voltage.

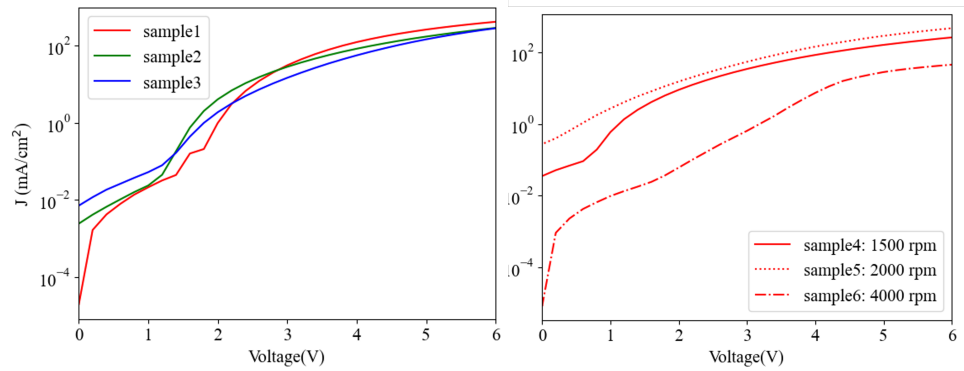


Figure 2.22: The J-V characterization of the devices.

the lowest current density, it may be due to the leakage current of the device.

d. EQE

EQE under different current density is shown in Fig. 2.23.

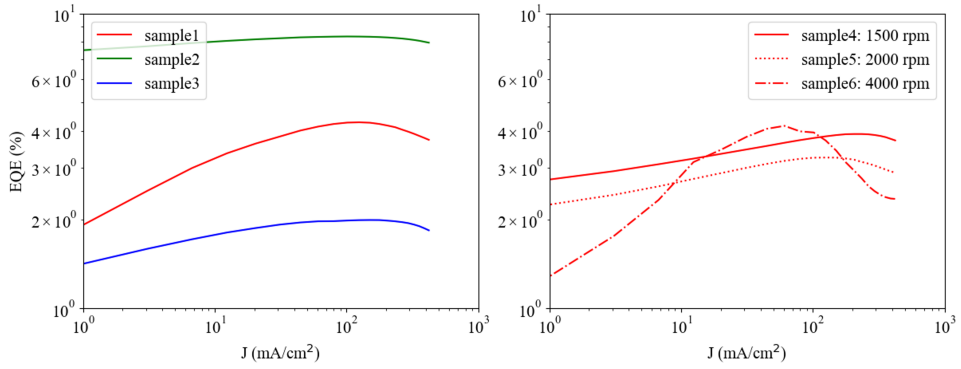


Figure 2.23: EQE of device under different luminance.

It can be observed that G-QD has the highest EQE, and the B-QD has the lowest EQE. The EQE of the devices with different ETL thickness is shown in Fig. 2.23(b).

2.5 Alternating Current Electroluminance Device

2.5.1 Introduction

We have already introduced the OLED and QLED devices in the previous sections. Both of these devices are driven by direct current (DC). They have high purity of color, brightness and efficiency. But the are also they also face the problem of cost and lifetime. Alternating current electroluminescent (ACEL),a type of electronic display technology that utilizes thin films of luminescent materials to produce light. Different from the devices fabricated in the previous experiments, the ACEL devices are driven by alternating current (AC). It is much easier to fabricate and more budget-friendly compared with the OLED or QLED.

Device Structure

The structure of device is given in Fig. 2.24.

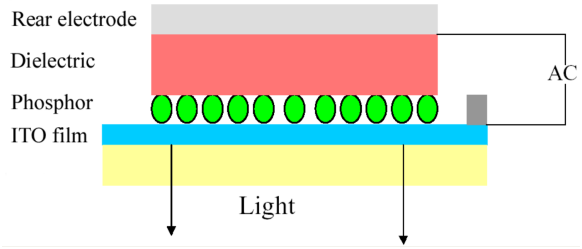


Figure 2.24: Device structure.

It contains serval layers, including the ITO substrate, phosphor layer, dielectric layer, and the electrode. The phosphor layer is the key component of the ACEL devices. When AC voltage is applied across the electrodes, the electrons in the phosphor layer are excited. Before the shift of the electric field, a polarized electric field is created. The reverse electric field will enhance the polarized electric field, and increase the luminance of the device.

Materials and its function

The following materials were used to fabricate the ACEL devices:

1. **ZnS:Cu** - Phosphor layer, a luminescent material composed of zinc sulfide (ZnS) doped with copper (Cu) ions. The copper ions introduce an impurity energy level in the ZnS bandgap, allowing the material to emit light when excited by an electric field.

2. **$BaTiO_3$** - Dielectric layer, a ceramic material with high dielectric constant and low dielectric loss. The $BaTiO_3$ layer is used to store and release energy in response to the applied electric field, enhancing the luminance of the device. It can also avoid direct contact between the phosphor layer and the electrode, which may cause a short circuit of the device.

3. **Ag** - Electrode, a conductive material used to apply the AC voltage across the device.

Fabrication Process

The ITO layer was deposited on the glass substrate before the experiment. Then, the substrate was held on a tooling system and was covered with the screen. The pattern of the screen was shown in Fig. 2.25 The first material which is printed on the substrate

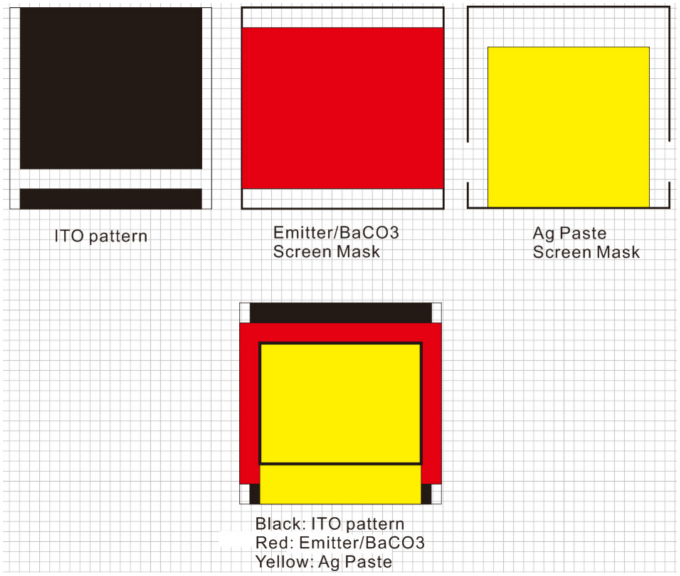


Figure 2.25: Screen mask design.

is ZnS:Cu phosphor. Then, the device was placed on a 120 °C heating plate for 10 min to evaporate the solvent. After the phosphor was printed on the screen, the screen was changed and was going to print the $BaTiO_3$ dielectric layer. For this layer, the mask was slightly larger than the previous one. The printing process was repeated three times to ensure the $BaTiO_3$ layer was fully covered. After that, the device was also placed on the heating plate for 120 °C and 10 min to evaporate the solvent. Before printing the Ag electrode, the extra phosphor and $BaTiO_3$ which was printed on the separated ITO at the bottom of the device, needed to be cleaned manually. At last, the Ag was printed on the device and placed on the heating plate of 120 °C and 10 min.

2.5.2 Results and Discussion

a. Spectrum

The spectrum of the devices under different AC frequencies is shown in Fig. 2.26. It can be observed that the optical appearance of ACEL will change from blue to green with the increase of the AC frequency. It indicates that the energy supplied to the Cu ions decreases with the rise of the AC frequency.

And the spectrum under different peak-to-peak voltage (Vpp) is shown in Fig. 2.27. It can be observed that the spectrum becomes more concentrated with the increase of the Vpp.

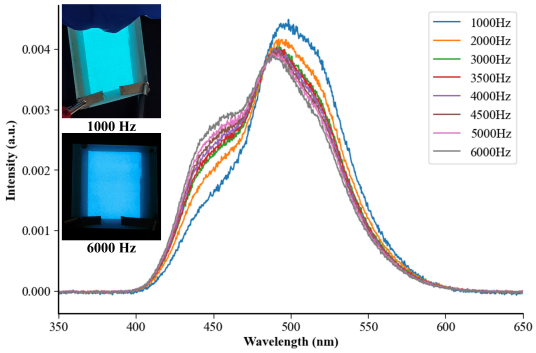


Figure 2.26: Spectrum of ACEL under different frequency.

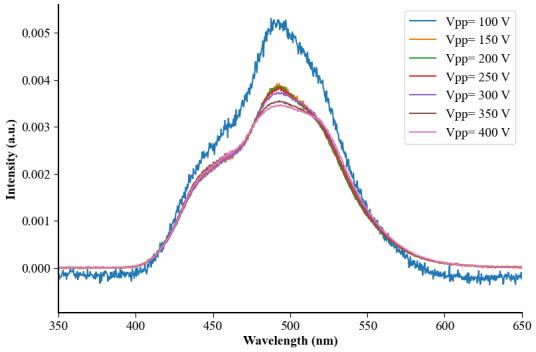


Figure 2.27: Spectrum of ACEL under different Vpp.

b. Luminous

The luminance of the devices under different AC frequencies is shown in Fig. 2.28. It can be observed that the luminance increases continuously from 100 Hz to 3000 Hz. The luminance decreases when the AC frequency is higher than 3000 Hz. This phenomenon is similar to the resonance phenomenon; when the frequency matches the "natural frequency" of the device, the luminance will reach the maximum. When the frequency is too high or too low, it will limit the energy supplied to the Cu ions, decreasing luminance.

The luminance of the devices with 3000 Hz AC frequency under different voltage is shown in Fig. 2.29. The luminance increases continuously from Vpp = 20 V to 300 V. Due to the limitation of the device, the Vpp can no longer increase, and the limitation of the device is not clear in this experiment. Theoretically, when the Vpp reaches a certain limit, the luminance will no longer increase, as the energy supplied to the Cu ions is enough to excite the electrons in the phosphor layer.

c. Customized patterning

Another advantage of the screen printing technique is that it can easily change the shape of the lighted area by changing the screen mask. The pattern of the lighted area is shown in Fig. 2.30. Three famous animation characters are printed on the device. In conclusion, ACEL and screen printing technology can be used for stylish and personalized applications.

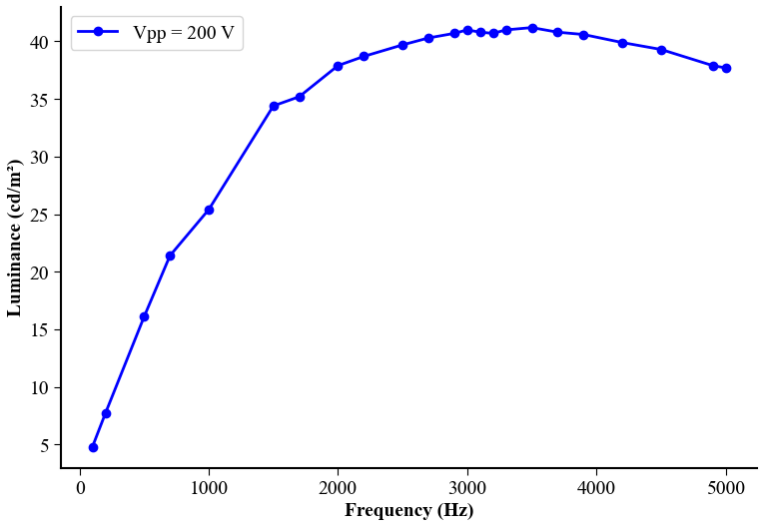


Figure 2.28: Luminous of ACEL under different frequency.

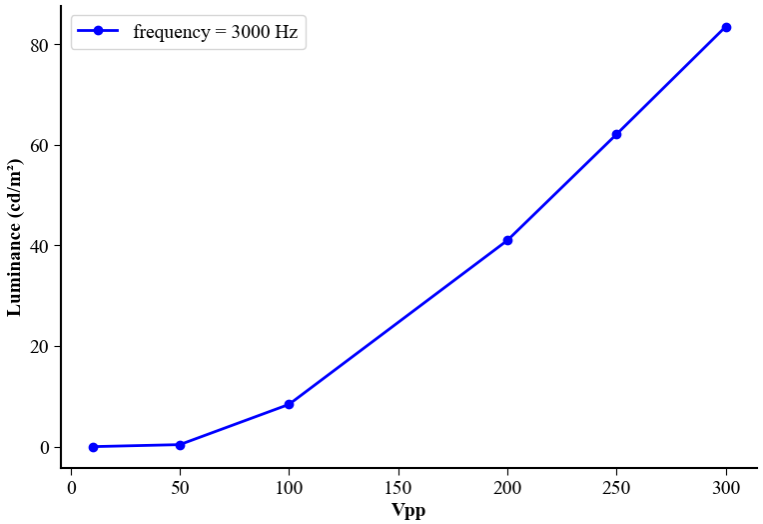
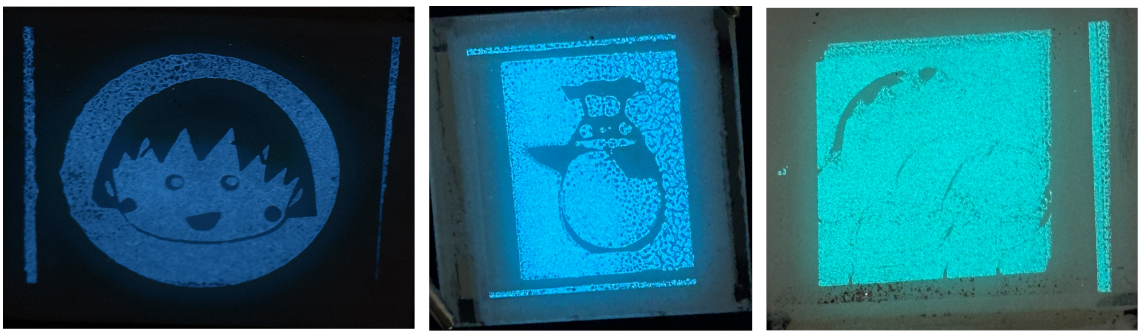


Figure 2.29: Luminous of ACEL under different Vpp.



(a) Chibi Maruko-chan **(b) Totoro** **(c) Crayon Shin-chan**

Figure 2.30: Pattern of the lighted area.

2.6 Organic Photovoltaic Device

2.6.1 Introduction

We have already introduced a lot of light-emitting devices in the previous sections. The energy can not only transfer from electricity to light but also from light to electricity. The organic photovoltaic device (OPV) is a type of solar cell that utilizes organic materials to convert sunlight into electricity. OPVs offer several advantages over traditional silicon-based solar cells, including flexibility, lightweight, and low-cost manufacturing.

The working principle of OPVs is based on the photo-induced charge transfer between a donor material and an acceptor material, leading to the formation of excitons, which then dissociate into free electrons and holes that are collected by electrodes to generate electrical current. Due to the unique properties of OPVs, they have attracted significant attention in recent years for applications in portable electronics, building-integrated photovoltaics, and other areas.

Device Structure

The structure of device is given in Fig. 2.31. It contains serval layers, including the ITO substrate, the charge selective layer, the photoactive layer, and the cathode. The photoactive layer is the most important layer in the device, which is responsible for the conversion of light into electricity.

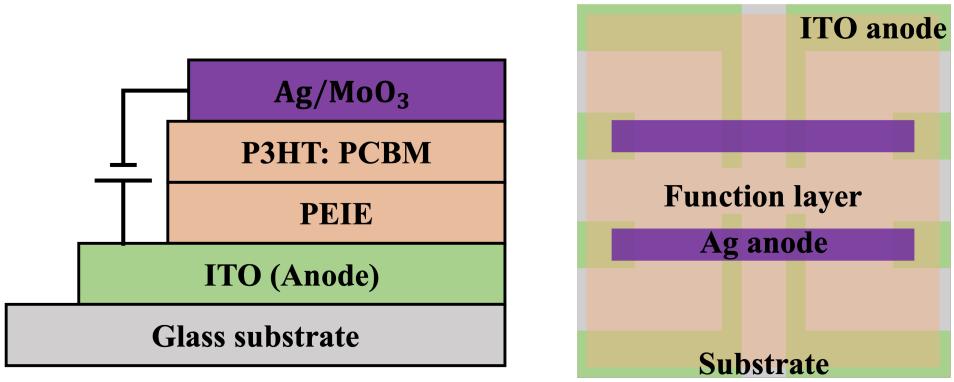


Figure 2.31: Device structure.

The thickness of the photoactive layer is a critical parameter that affects the performance of the device. The active part of the cell is limited by the exciton diffusion length. Instead of using two separate layers, mixing the donor and acceptor materials together to form a bulk heterojunction can increase the exciton dissociation efficiency.

The energy band diagram of the heterojunction organic solar cell is shown in Fig. 2.32.

It needs several stages to transfer the photon energy into electrical energy:

1. **Photon Absorption:** The photoactive layer absorbs photons and generates excitons.

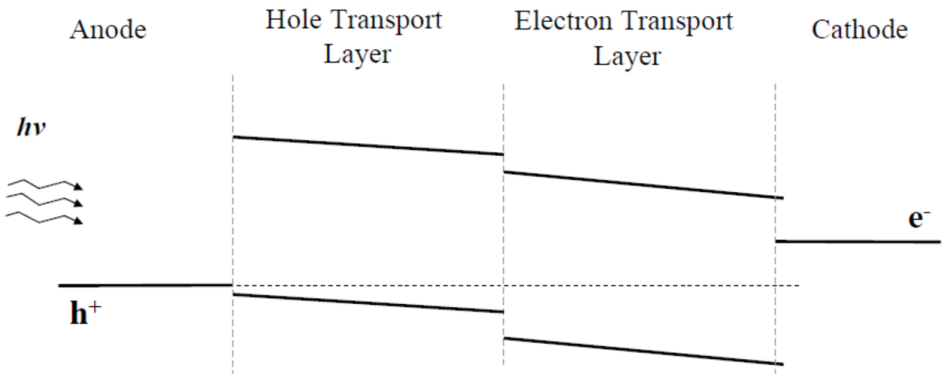


Figure 2.32: Energy band diagram of the heterojunction organic solar cell.

2. **Exciton Formation:** The excitons are formed in the photoactive layer.
3. **Exciton Diffusion:** The excitons diffuse to the donor-acceptor interface.
4. **Exciton Dissociation:** The excitons dissociate into free electrons and holes at the donor-acceptor interface.
5. **EHP Formation:** The free electrons and holes form electron-hole pairs (EHPs).
6. **Charge Collection:** The electrons and holes are collected by the electrodes to generate electrical current.

Materials and its function

The following materials were used to fabricate the ACEL devices:

1. **PEIE** or Polyethyleneimine Ethoxylated, is employed as an interfacial layer between the photoactive layer and the electrode, particularly at the cathode. It acts as a charge selective layer, allowing electrons to pass through while blocking holes. And it can enhance the stability of the device at the same time.
2. **P3HT: PCBM** or Poly(3-hexylthiophene) and [6,6]-phenyl-C61-butyric acid methyl ester, is a blend of two materials that are used as the photoactive layer in the device. P3HT is a classic donor material, and PCBM is a classic acceptor material. The blend of the two materials is used to create a bulk heterojunction, which is a key component of OPV devices.

Fabrication Process

The ITO layer was deposited on the glass substrate before the experiment. Then, 110 ul, 0.8% PEIE dissolved in 2-methoxyethanol was coated on the device using spin coating speed of 5000 rpm and 40 s. After spin coating, the device was placed on a heating plate of 100°C for 5 min. The spin coating of P3HT: PCBM followed the similar sequence, but under different spin speed. 100 ul P3HT: PCBM (1:1, 17 mg/ml, dissolved in ODCB) was coated on the devices with 800, 1400, and 2000 rpm, respectively. The backing condition

is 120°C and 10 min. The deposition of 8 nm MoO3 and Ag using vacuum evaporation was done after P3HT: PCBM was coated on the device.

Table 2.5: Fabrication settings of different devices.

Device	PEIE	P3HT: PCBM	Spin Speed (rpm)
S1	0.8%	0.05%	800
S2	0.8%	0.05%	1400
S3	0.8%	0.05%	2000
S4	0.01%	0.05%	800
S5	0.8%	0.05%(Without electrode)	800
S6	0.8%	0.05%(Without electrode)	1400
S7	0.8%	0.05%(Without electrode)	2000

2.6.2 Result and Discussion

a. Transmittance

Samples S5-S7 were fabricated without the electrode, which is used to discuss the transmittance of the active layer. To visualize the transmittance more clearly, a moving average filter ([WIN_SIZE=10]) was applied to the data. The transmittance of the devices under different wavelength is shown in Fig. 2.33 below. Transmittance is a key value to estimate the absorption spectrum of the OPV devices. The transmittance spectrum indicates that P3HT: PCBM has a absorption of light from 450 to 650 nm, the highest absorptivity is located at 500 nm. The thicker active layer which is coated under slower spin speed has a higher absorptivity. However, the thickness of the active not only affects the absorptivity but also affects the electronic property of OPV which will be discussed in the following chapter.

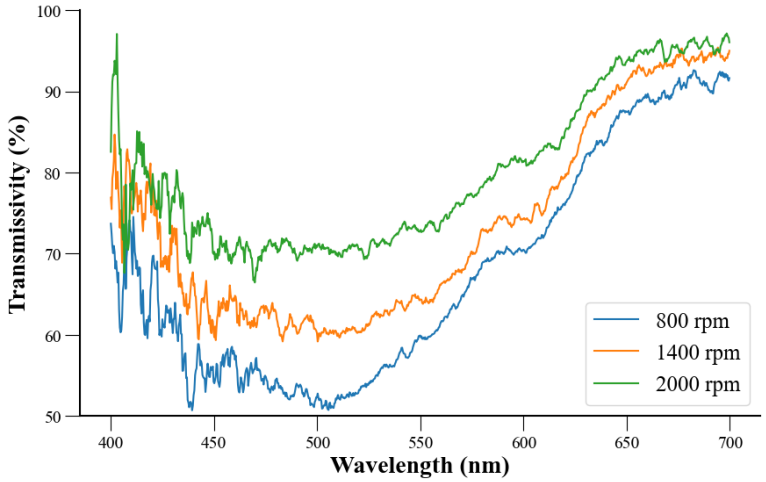


Figure 2.33: The transmittance of the active layer of the OPV devices.

b. Electronic Properties

The electronic properties of OPV was measured by measuring the dark and bright J-V curves of OPV. The light J-V curves was measured by putting OPV under a AM1.5G sunlight simulator which has a power density of 1000 W/m^2. The light current density is shown in Fig. 2.34.

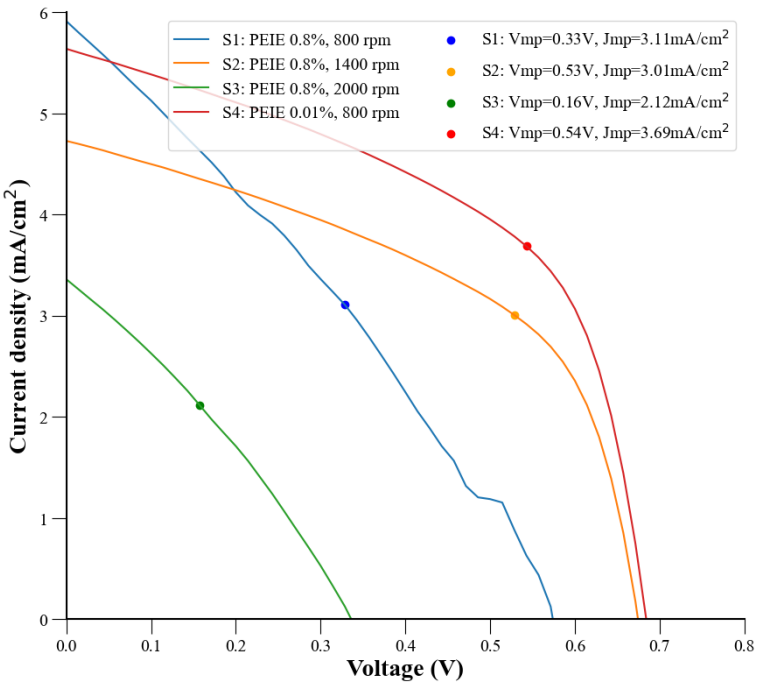


Figure 2.34: The light current density of the OPV devices.

The maximum power point, P_{mpp} refers to the point where the power output of the device is maximized. At this point the current and voltage are defined as J_{mpp} and V_{mpp} respectively. It can also be calculated by Eq. (2.8). And the maximum power point of these four devices were marked in Fig. **??** respectively.

$$P_{mpp} = J_{\max} \times V_{\max} \tag{2.8}$$

Fill factor (FF) is a key parameter to evaluate the performance of the OPV devices. It is defined as the ratio of the maximum power to the product of the open-circuit voltage and the short-circuit current. The fill factor is calculated by the following equation:

$$FF = \frac{P_{mpp}}{J_{\text{sc}} \times V_{\text{oc}}} \tag{2.9}$$

The power conversion efficiency (PCE) can be calculated by dividing the product of the short-circuit current, open-circuit voltage, fill factor, and the power of the incident light. The PCE is calculated by Eq. (2.10).

$$PCE = \frac{J_{\text{sc}} \times V_{\text{oc}} \times FF}{\text{Power of Incident Light}} \tag{2.10}$$

c. Absorption Efficiency

The absorption efficiency of the active layer is given by Eq. (2.11).

$$\frac{\int A \times AM1.5GSolarSpectrum}{1000W/m^2} \tag{2.11}$$

A is the absorptivity of the activity layer. And the electronic performance of these devices is shown in Table 2.6.

Table 2.6: Electronic performance of the OPV devices.

Device	Short-circuit Current J_{sc} (mA)	Open-circuit Voltage V_{oc} (V)	FF	PCE
S1	5.91	0.57	30.1%	1.01%
S2	4.73	0.67	49.8%	1.58%
S3	3.10	0.28	29.6%	0.26%
S4	5.64	0.68	51.9%	1.99%

Chapter 3

Summary and Conclusion

The Optoelectronic Devices Fabrication Laboratory course offers me a comprehensive learning experience in the fabrication and characterization of optoelectronic devices. This course provides a hands-on approach to understanding the fundamental principles and practical techniques involved in the fabrication of devices such as OLEDs, QLEDs, ACEL, OPV and other optoelectronic devices.

Throughout the curriculum, I have had the opportunity to explore various thin-film deposition methods, including thermal evaporation, sputtering, and spin-coating. The course has emphasized the importance of precise control over film thickness and composition in order to engineer devices with specific properties. Additionally, I have learned about patterning technologies such as lithography, laser direct writing, and screen printing, which are essential for creating device architectures and intricate patterns.

One of the highlights of the course has been the inclusion of advanced characterization tools and techniques. I have gained hands-on experience in using optical spectroscopy to study the light interaction characteristics of optoelectronic devices through transmission, reflection, and absorption measurements. Furthermore, I have learned about electrical assessment methods, including the four-probe technique, which have allowed me to measure parameters such as resistivity and conductivity.

By actively engaging in the fabrication and characterization processes, I develop a deep understanding of the underlying principles and challenges involved in optoelectronic device fabrication. I acquire valuable skills that can be applied in research and industry settings, contributing to advancements in optoelectronic technologies.

In conclusion, the Optoelectronic Devices Fabrication Laboratory course has been a transformative educational experience. It has equipped me with a strong skill set and a profound knowledge base that are essential for making significant contributions to the development of cutting-edge optoelectronic technologies and for pursuing a successful career in this dynamic field.